

GeoAdapt: Few-shot Peg Insertion Adaptation through Aligned Geometric Observations

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Abstract: Robotic insertion tasks require precise geometric alignment despite severe occlusion and contact-rich interactions. Although recent learning-based insertion methods have achieved strong performance on standard peg insertion tasks, they often struggle to adapt to unseen insertion geometries that require precise rotational alignment. A key limitation is that existing insertion datasets primarily contain simple and highly regular insertion interfaces, providing limited diversity in alignment behaviors. To address this challenge, we present **GeoAdapt**, a framework for few-shot insertion adaptation based on procedurally generated insertion geometries and aligned geometric observations. First, we introduce a procedural generation pipeline for constructing diverse plug-socket geometries with varying rotational alignment constraints, substantially increasing geometric diversity at the insertion interface. Second, we propose an Aligned Geometric Observation Space (AGOS) that preserves rotational alignment information under severe occlusion through virtual geometric alignment between plug and socket observations. AGOS additionally enables physically consistent rotational augmentation for improved few-shot adaptation. Finally, we incorporate force-conditioned policy learning and recovery-oriented demonstrations to improve robustness during contact-rich insertion. Through simulation and real world experiments, we demonstrate strong zero-shot generalization and few-shot adaptation to unseen insertion geometries with challenging rotational alignment requirements.

Keywords: Robotic Insertion, Few-shot Adaptation, Diffusion Policy, Contact-rich Manipulation, Geometric Representation

1 Introduction

Insertion and assembly are fundamental robotic manipulation problems that require precise geometric alignment under contact-rich interactions. These tasks arise in a wide range of industrial and domestic settings, where robots must reliably align and insert objects despite tight tolerances, partial observability, and physical contact. Recent learning-based approaches have demonstrated promising performance on peg insertion tasks using visual and force feedback [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14]. However, existing insertion benchmarks and datasets primarily focus on relatively simple insertion geometries, such as circular holes or basic rectangular connectors [1, 15, 16]. In contrast, many real-world insertion tasks involve complex and asymmetric plug-and-socket geometries that require precise rotational alignment.

This geometric gap presents a major challenge for robotic adaptation. In practical deployment scenarios, robots must rapidly adapt to previously unseen insertion geometries, either zero-shot or from only a small number of demonstrations. However, policies trained on existing insertion datasets often fail to generalize to novel geometries with different rotational alignment constraints and contact behaviors. In this work, we explore *how to enable robotic policies to rapidly adapt to novel complex insertion geometries from limited demonstrations*.

Fast adaptation for complex insertion geometries presents four primary challenges. First, insertion suffers from severe visual occlusion. As the plug approaches the socket opening, the plug increasingly occludes the socket geometry, while the plug tip also becomes difficult to observe due to physical contact and limited camera visibility. This obscures critical visual cues when accurate corrections are most important. Second, high-precision insertion often has sub-millimeter error-tolerance under contact-rich interactions. Even small translational or rotational errors can easily cause the plug to get stuck around the socket opening, requiring active recovery behaviors through local corrective motions. Third, generalizable insertion policies must generalize not only to unseen geometries, but also to different insertion orientations and contact conditions. Finally, existing insertion datasets often provide limited diversity at the insertion interface itself. While prior datasets often contain diverse global object shapes, the plug and socket shapes around the insertion region frequently remain simple and regular geometries, limiting the diversity of alignment behaviors seen during training [1, 15, 16].

To address these challenges, we propose **GeoAdapt**, a framework for few-shot insertion adaptation through procedurally generated insertion geometries and aligned geometric observations. First, we introduce a procedural generation pipeline for constructing diverse plug-and-socket geometries with varying rotational alignment constraints, substantially increasing geometric diversity at the insertion interface. Second, we propose an **Aligned Geometric Observation Space (AGOS)** that preserves plug-socket alignment information under severe occlusion by constructing aligned virtual geometric observations. AGOS additionally enables physically consistent rotational augmentation, substantially improving adaptation from limited demonstrations. Finally, we incorporate force-conditioned policy learning together with recovery-oriented demonstrations to improve robustness during contact-rich insertion failures. Force feedback provides additional signals for corrective behaviors when visual observations become ambiguous due to occlusion and contact [2, 3, 4, 5, 6, 7, 8, 11]. We experimentally show that, by combining aligned virtual observations and force-conditioned policy learning, our approach enables stronger insertion adaptation under severe occlusion and precision constraints.

In summary, this paper makes the following core contributions:

1. We introduce a method for creating a procedurally generated industrial insertion dataset featuring diverse plug and socket geometries with strict rotational constraints, significantly expanding beyond traditional peg-in-hole benchmarks.
2. We propose an aligned virtual observation representation that preserves geometric observability under severe occlusion and enables physically consistent rotational data augmentation for few-shot adaptation.
3. We develop a multimodal framework integrating aligned virtual observations with force-conditioned policy learning, demonstrating strong few-shot adaptation and recovery behaviors on challenging, unseen insertion geometries.

2 Related Works

2.1 Datasets for Robotic Assembly

Numerous asset datasets and asset-generation methods are commonly used for robotic manipulation [17, 18, 19, 20, 21, 22, 23]. Some of these datasets contain a large number of CAD assets [20, 21]. Meanwhile, another line of work studies 3D asset generation using generative models [24, 25, 26, 27, 28, 29, 30, 31]. However, most of those existing datasets mainly contain a single asset rather than paired assets for insertion tasks. More recent methods generate paired assembly datasets or assembly-compatible geometries by generating the matching pair of a given asset [1, 15, 16]. Despite the diverse global appearance of plugs and sockets, the insertion interfaces themselves often remain relatively regular geometries, such as square or circular holes. Our work is the first to procedurally generate insertion interfaces with more irregular, asymmetric and complex geometries for improving policy generalization to novel insertion tasks requiring challenging rotational alignment.

88 2.2 Adaptation and Generalization for Robotic Insertion

89 Several works study adaptation for robotic manipulation through in-context imitation learning, key-
90 point correspondence, graph-based representations, and adaptable motion priors [32, 33, 34, 35, 36,
91 37, 38]. These methods improve adaptation across object appearance and spatial relationships, but
92 do not explicitly address adaptation requiring precise alignment under severe occlusion in contact-
93 rich insertion. Some other works study data-efficient insertion learning and multisensory augmen-
94 tation for insertion tasks [6, 12]. However, these methods mainly focus on insertion tasks with
95 relatively regular insertion interfaces. In contrast, our work focuses on fast adaptation to unseen in-
96 sertion geometries involving complex rotational alignment. We find that aligned virtual observations
97 naturally provide a physically consistent augmentation space for rotational augmentation, improving
98 adaptation from limited demonstrations.

99 2.3 Synergizing Vision and Force for Contact-rich Insertion

100 A number of works combine visual observations and force feedback for contact-rich insertion un-
101 der occlusion [2, 3, 4, 5, 6, 7, 8, 11, 39]. While force feedback helps contact state estimation and
102 recovery, it provides limited information for precise rotational alignment on geometrically complex
103 insertion interfaces. In contrast, we propose aligned geometric observations that preserve rotational
104 alignment information under severe occlusion and improve adaptation to unseen insertion geome-
105 tries. Recent work has also explored tactile-based peg insertion by reconstructing local contact ge-
106 ometry from tactile observations and estimating relative peg-hole pose [14]. However, these methods
107 primarily focus on explicit pose estimation from tactile contacts rather than learning-based insertion
108 adaptation and recovery for unseen insertion geometries. Our work instead learns force-conditioned
109 insertion policies from aligned geometric observations and recovery-oriented demonstrations.

110 3 Problem Formulation

111 We study fast adaptation for top-down peg insertion tasks requiring precise translational and ro-
112 tational alignment between a plug and socket. Let $\mathcal{T}$ denote an insertion task with unseen plug
113 and socket geometries. Given an insertion policy π_θ pretrained on a large-scale insertion dataset
114 $\mathcal{D}_{\text{pre}}$, our objective is to adapt the policy to a novel task $\mathcal{T}_{\text{adapt}}$ using a small set of task-specific
115 demonstrations $\mathcal{D}_{\text{adapt}}$.

116 At each timestep t , the policy receives observations $o_t = \{I_t^w, f_t\} \in \mathcal{O}$, where I_t^w denotes the wrist-
117 camera depth observation and $f_t \in \mathbb{R}^3$ denotes the end-effector linear force observation. The policy
118 predicts delta end-effector actions $a_t = (\Delta p_t, \Delta r_t) \in \mathcal{A}$, where $\Delta p_t \in \mathbb{R}^3$ represent translation
119 actions in the wrist camera frame and $\Delta r_t \in \mathbb{R}^6$ represents rotational actions in the world frame.

120 We focus on the final contact-rich insertion stage and make the following assumptions: (1) the plug
121 is already grasped in a vertical configuration before insertion, (2) the plug does not slip relative to
122 the gripper during insertion, and (3) the socket remains fixed on the table during insertion. We focus
123 on tasks requiring high-precision translational as well as rotational alignment in the yaw direction,
124 which is the major rotational correction during top-down insertion.

125 4 Method

126 We first procedurally generate diverse insertion geometries for large-scale insertion pretraining (Sec-
127 tion 4.1). We then introduce an aligned geometric observation space that preserves rotational align-
128 ment information under severe occlusion (Section 4.2). Finally, we present our force-conditioned
129 diffusion policy and recovery-oriented adaptation framework for few-shot transfer to unseen inser-
130 tion geometries (Sections 4.3–4.5).

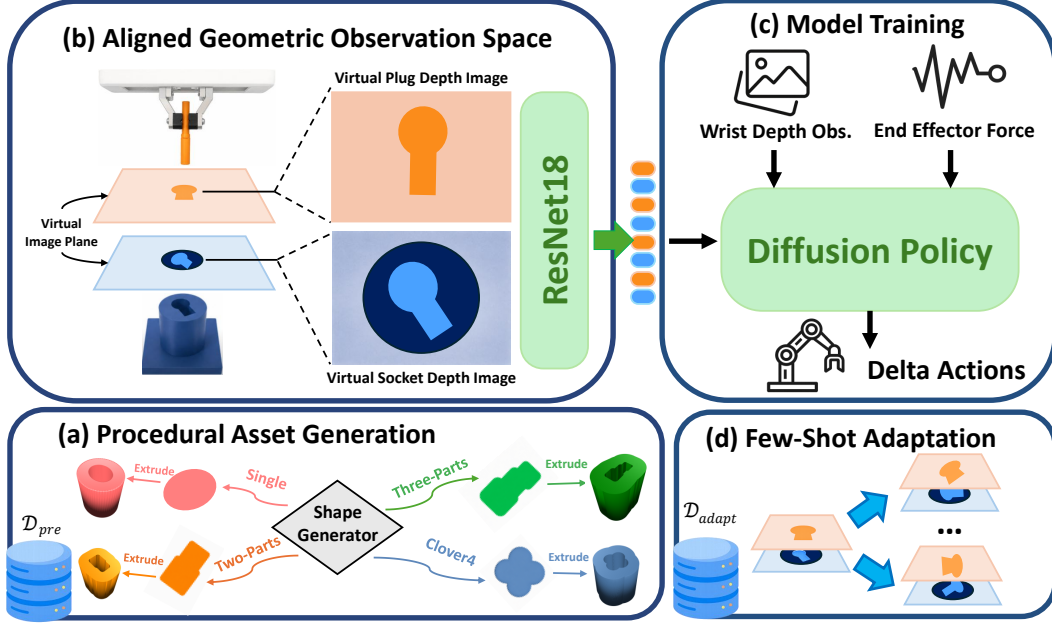


Figure 1: System overview of **GeoAdapt**. (a) Diverse plug–socket geometries are procedurally generated to construct the pre-training dataset $\mathcal{D}_{pre}$. (b) The proposed aligned geometric observation space (AGOS) projects plug and socket geometries onto virtual image planes to produce canonical virtual depth observations. (c) A diffusion policy is trained using wrist depth observations, AGOS representations, and end-effector force feedback to predict delta actions. (d) During few-shot adaptation, a small adaptation dataset $\mathcal{D}_{adapt}$ is used together with rotational and translational augmentation in AGOS to adapt the policy to unseen insertion geometries.

131 4.1 Procedural Generation of Insertion Geometries

132 Existing insertion datasets mainly contain insertion interfaces with relatively regular geometries,
 133 limiting geometric diversity during training [6, 12]. As a result, policies trained on these datasets
 134 often struggle to generalize to unseen insertion tasks requiring precise rotational alignment. To
 135 address this limitation, we procedurally generate diverse plug–socket geometries designed to induce
 136 a broad range of rotational constraints and contact interactions.

137 Our generation pipeline (illustrated in Fig. 1a) constructs insertion interfaces by composing multiple
 138 geometric primitives, including circles, ellipses, rectangles, and rounded rectangles, with indepen-
 139 dently randomized scale, position, and orientation. We vary the structural layout of the insertion
 140 interface by arranging these components into different configurations, such as single-component,
 141 multi-component, and clover-shaped structures. These layouts induce different rotational symme-
 142 tries and alignment constraints, substantially increasing the diversity of insertion behaviors encoun-
 143 tered during training. By perturbing the relative offsets and orientations of these primitives, we
 144 intentionally reduce rotational symmetry and generate asymmetric insertion interfaces with varying
 145 alignment difficulty. See Appendix C for shapes that we can generate.

146 The procedural generation process proceeds as follows: First, multiple primitive components are
 147 sampled and transformed in 2D. The transformed primitives are then merged into a unified insertion
 148 interface contour. We subsequently extrude the resulting interface geometry to construct paired plug
 149 and socket meshes. Finally, we apply insertion clearance constraints to ensure physically feasible
 150 insertion interactions while preserving geometric diversity.

151 This procedural generation strategy substantially expands the diversity of insertion interfaces be-
 152 yond traditional peg-in-hole benchmarks. Rather than only varying global object appearance, our

approach directly increases diversity at the insertion interface itself, exposing policies to a wider distribution of rotational alignment relationships and contact behaviors during training.

4.2 Aligned Geometric Observation Space (AGOS)

Rotational alignment information becomes increasingly difficult to observe during insertion due to severe occlusion. As the plug approaches the socket opening, the plug progressively occludes the socket geometry, while the plug tip itself becomes difficult to observe because of physical contact and limited camera visibility. As a result, policies that directly reason over wrist-camera observations may lose critical geometric alignment information precisely when accurate corrections are most important.

To address this challenge, we construct an occlusion-aware aligned geometric representation that explicitly preserves the relative geometric relationship between the plug and socket throughout insertion. To collect the data needed for this representation, we use a wrist-mounted depth camera together with an auxiliary depth camera placed beneath the insertion workspace. Before insertion, the robot moves the grasped plug above the auxiliary camera while maintaining the plug’s orientation. The auxiliary camera captures a detailed depth image of the plug geometry. Separately, the wrist camera captures a detailed depth image of the socket geometry before insertion.

We use these detailed plug and socket images to construct an **Aligned Geometric Observation Space (AGOS)** that contains the important visual details about how the plug and socket are aligned, even during the contact-rich portions of insertion when there is heavy occlusion. AGOS explicitly represents the relative geometric relationship between the plug and socket in a shared aligned frame (see Fig. 1b and d).

To construct AGOS, we first acquire geometric observations of both the plug and socket before insertion. The socket depth observation is converted into a point cloud and projected into a virtual top-down view. The plug geometry is similarly projected into a virtual image plane with matched camera intrinsics, producing plug and socket observations in a shared virtual frame. See Appendix D for detailed implementations.

During insertion, we continuously update the virtual socket image according to the current gripper yaw rotation, such that the plug and socket images remain geometrically aligned throughout insertion. Specifically, let ψ_t denote the yaw rotation of the gripper relative to the initial insertion pose. We rotate the aligned socket depth observation in the opposite direction: $\hat{I}_s^t = R(-\psi_t) I_s^0$, where $I_s^0 \in \mathbb{R}^{H \times W}$ is the initial socket depth image and $R(\cdot)$ denotes image-space rotation. Because the plug observation is captured from an upward-facing camera while the socket observation is represented in a top-down virtual frame, we flip the plug depth image to align the two geometries. The plug observation is acquired only once before insertion begins and remains fixed throughout the insertion process; only the socket observation is updated according to the current gripper orientation.

Finally, we must apply geometric scale calibration to align the plug and socket geometry scales. Let us denote $\tilde{I}_p$ as the scaled aligned plug depth image. The resulting Aligned Geometric Observation Space (AGOS) is given by $\mathcal{G}_t = \{\tilde{I}_p, \hat{I}_s^t\}$.

The resulting aligned geometric observation space preserves plug-socket alignment information throughout insertion despite severe occlusion. With the Aligned Geometric Observation Space (AGOS), rotational alignment reduces to reasoning about the relative rotation required to align the plug and socket geometries, i.e. aligning $\hat{I}_s^t$ with $\tilde{I}_p$. Furthermore, because both plug and socket observations are represented in a shared aligned frame, we can apply a joint rotational augmentation to the aligned observations while preserving the underlying geometric relationships. This substantially improves few-shot adaptation to unseen insertion geometries.

4.3 Combining the Force Modality

While AGOS preserves geometric alignment information under visual occlusion, small translational or rotational errors can cause the plug to become stuck near the insertion opening. In these cases,

visual observations alone may be insufficient to infer the contact state. We therefore incorporate end-effector force feedback as an additional observation modality.

At each timestep, we measure the linear force observation $f_t \in \mathbb{R}^3$ at the robot end effector. Rather than directly using raw force observations, we compute a residual force relative to the initial insertion state: $\Delta f_t = f_t - f_0$, where f_0 denotes the force measurement before insertion begins. This normalization removes force components caused by gravity and initial grasp contact, allowing the policy to focus on force changes caused by contact interactions between the plug and socket.

To align the force observation with the visual observation frame, we transform the force observation into the wrist camera frame: $\hat{f}_t = R_{cw}\Delta f_t$, where R_{cw} denotes the rotation from the robot end effector frame to the wrist camera frame. This transformation aligns the force modality with the wrist-camera observation space, enabling more consistent geometric reasoning between visual observations and force contact signals.

4.4 Policy Architecture

We build our policy on top of Diffusion Policy [40]. The policy predicts delta end-effector actions $a_t = (\Delta p_t, \Delta r_t)$, where $\Delta p_t \in \mathbb{R}^3$ denotes the translation actions in the wrist camera frame, and $\Delta r_t \in \mathbb{R}^6$ denote the rotational actions represented in the robot base frame using the 6D rotation representation [41].

Our policy takes three observation modalities as input: wrist depth observations I_t^w , aligned geometric observations $\mathcal{G}_t$, and aligned force observations $\hat{f}_t$. The wrist depth observation is encoded using a single-channel ResNet18 encoder [42]. The aligned geometric observation space $\mathcal{G}_t = \{\tilde{I}_p, \hat{I}_s^t\}$ is stacked into a two-channel depth representation and encoded using a separate ResNet18 encoder [42]. The aligned force observation is processed using a multilayer perceptron.

The latent features from all modalities are concatenated and used to condition the diffusion policy. During inference, the policy predicts action chunks of length 10, and the first 5 actions are executed.

4.5 Few-shot Adaptation via Geometric Augmentation

As illustrated in Figure 1(d), AGOS enables rotational augmentation by applying the same rotation to both the aligned plug and socket observations. Because the augmentation preserves the underlying plug-socket geometric relationship represented in AGOS, the corresponding action and force labels can be reused without modification. This augmentation increases rotational diversity and improves data efficiency during few-shot adaptation. See Appendix E for our data trajectory collection algorithm.

5 Experiments

We structure our experimental evaluation around the following three research questions:

RQ-1: Generalization and Adaptation to Unseen Geometries. Can our method **GeoAdapt** achieve stronger zero-shot generalization and few-shot adaptation to unseen insertion geometries involving challenging rotational alignment?

RQ-2: Aligned Geometric Observation Space (AGOS) Design. Does AGOS help insertion? Does the geometric scale calibration and rotational augmentation in the aligned geometric observations contribute to insertion adaptation performance?

RQ-3: Real-world Transfer. Does GeoAdapt work for insertion tasks in the real world with a real robot?

Table 1: Adaptation Success Rates on Unseen Geometries. Tasks are ordered by increasing rotational alignment difficulty from left to right. We report success rates (%) on eight unseen socket and plug geometries in simulation, evaluated over 100 rollouts per geometry. GeoAdapt achieves the highest average success rate compared to its ablated variants and the GeoAdapt-Automate baseline.

Methods									Average
GeoAdapt	90	86	69	83	61	65	49	53	69.5
w/o Force	72	89	62	59	52	59	44	45	60.3
w/o AGOS	78	72	66	71	55	68	29	42	60.1
w/o Scale Calibration	77	70	70	79	63	25	28	44	57.0
w/o Rot. Augmentation	92	82	73	66	55	49	37	53	63.4
GeoAdapt-Automate	68	32	26	20	31	5	11	8	25.1

5.1 Experimental Setup

Tasks. We select eight unseen insertion tasks generated by our procedural shape generator for evaluation. The geometries used in these tasks are excluded from the training dataset. All tasks require insertion within a clearance tolerance of 0.8 mm, making successful adaptation sensitive to geometric discrepancies.

Baselines. We compare our method **GeoAdapt** against five ablated variants: (1) **GeoAdapt w/o Force**, which removes force conditioning from the policy; (2) **GeoAdapt w/o AGOS**, which removes the aligned geometric observation space; (3) **GeoAdapt-Automate**, which trains the policy only on the Automate dataset without procedurally generated insertion geometries; (4) **GeoAdapt w/o Scale Calibration**, which removes the plug-socket geometric scale alignment step in AGOS construction; and (5) **GeoAdapt w/o Rotational Augmentation**, which removes the AGOS-based rotational augmentation used during pretraining and adaptation.

Datasets. Our pretraining dataset consists of both the Automate dataset [1] and our procedurally generated insertion dataset. For the Automate dataset, we filter out non-insertion tasks and only retain tasks involving plug-socket insertion, resulting in 67 insertion tasks used for pretraining. For our procedurally generated dataset, we generate 200 insertion tasks. For each task, we collect 72 demonstration trajectories. Among them, 24 are smooth insertion trajectories and 48 are recovery-oriented trajectories involving rotational and translational error recovery. For few-shot adaptation, we collect 20 task-specific demonstrations for each unseen task. Among them, 8 are smooth insertion trajectories and 12 are recovery-oriented trajectories.

Evaluation. We use insertion success rate as the evaluation metric for both simulation and real-world experiments. In simulation, each method is evaluated using 100 trials. We report both zero-shot success rates as well as adaptation success rates after fine-tuning on demonstrations of the unseen task.

5.2 Adaptation to Unseen Insertion Geometries (RQ-1)

Table 1 and Table 4 present zero-shot generalization and few-shot adaptation performance on unseen insertion geometries. The results indicate three key findings: (1) our **GeoAdapt** achieves stronger zero-shot generalization and few-shot adaptation performance on challenging rotational alignment tasks compared to all ablations. (2) procedural generation of diverse insertion interfaces significantly improves adaptation to unseen insertion geometries, whereas training only on the Automate dataset generalizes poorly to tasks requiring precise rotational alignment; and (3) both force conditioning and the AGOS are important for contact-rich insertion adaptation under severe occlusion.

Table 2: Real-world adaptation success rates on unseen insertion tasks. Each tested with 15 trials.

Task	GeoAdapt	GeoAdapt w/o AGOS	GeoAdapt-Automate
Task 1	13 / 15	9 / 15	11 / 15
Task 2	10 / 15	7 / 15	1 / 15
Task 3	9 / 15	6 / 15	0 / 15
Average	10.7 / 15	7.3 / 15	4 / 15

5.3 Aligned Geometric Observation Space Design (RQ-2)

Table 1 presents ablations on two components of the aligned geometric observation space: geometric scale calibration and rotational augmentation. Specifically, we compare the full **GeoAdapt** against variants that remove geometric scale calibration or rotational augmentation during pretraining and adaptation. The results indicate two key findings: (1) geometric scale calibration is important for learning consistent rotational alignment relationships between plug and socket observations; and (2) rotational augmentation is beneficial during few-shot adaptation with limited demonstrations.

5.4 Real-world Transfer (RQ-3)

Real-world Setup. We evaluate real-world adaptation using a Franka Panda robot equipped with a wrist-mounted ZED Mini depth camera and an auxiliary ZED mini depth camera placed orthogonally above the table, as shown in Figure 6. The auxiliary camera captures stable bottom-up plug depth observations for constructing the aligned geometric observation space. We 3D print three unseen PLA plug-socket insertion tasks with varying difficulty, each with a tolerance of 0.8 millimeter. Table 2 shows the real-world adaptation results on three unseen insertion geometries. GeoAdapt consistently achieves the highest success rate, demonstrating successful transfer from simulation to the real world. The results highlight the importance of both AGOS and procedurally generated insertion geometries for adapting to unseen real-world insertion tasks with challenging rotational alignment requirements.

6 Conclusion

We present **GeoAdapt**, a diffusion-based framework for few-shot industrial insertion adaptation under severe occlusion and contact-rich interactions. Our approach combines procedurally generated insertion geometries, aligned geometric observations, aligned force observations, and recovery-oriented demonstrations to improve adaptation to unseen insertion interfaces with challenging rotational alignment requirements. Through simulation and real world experiments, we show that GeoAdapt achieves strong zero-shot generalization and efficient few-shot adaptation on geometrically challenging insertion tasks. More broadly, our results suggest that explicitly modeling rotational geometric relationships during both pretraining and adaptation can substantially improve robustness for contact-rich insertion. We believe procedural geometric pretraining and aligned geometric representations provide a promising direction toward scalable adaptation of robotic insertion policies to increasingly diverse industrial geometries.

7 Limitations

While GeoAdapt demonstrates strong adaptation for complex insertion tasks, it presents a few limitations. First, constructing AGOS requires an auxiliary upward-facing camera and a preliminary scanning step, which may be impractical in highly cluttered or constrained workspaces. Second, our method assumes a rigid grasp where the plug does not slip relative to the gripper. Because the plug geometry is captured only once before insertion, any unobserved in-hand slip caused by contact forces will misalign the AGOS representation. Finally, our procedural generation and problem formulation primarily address top-down insertions of 2.5D extruded shapes, leaving unconstrained 6-DoF insertions of fully 3D interlocking parts as an important direction for future work.

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A Project Website

Anonymized Website: : <https://geoadaptinsertion.github.io/>

B Implementation Details

B.1 Policy Architecture

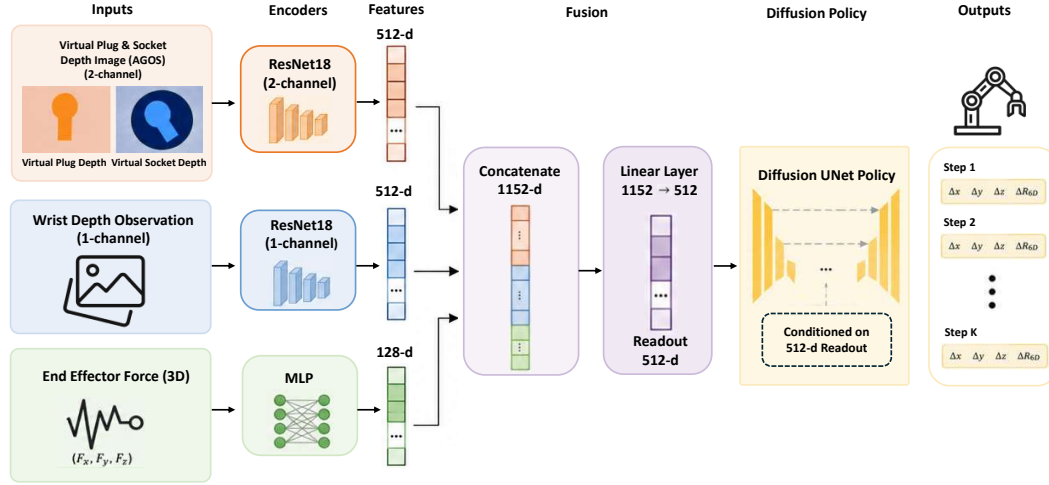


Figure 2: Overview of the GeoAdapt policy architecture. GeoAdapt takes three modalities as input: (1) a two-channel AGOS depth observation consisting of aligned virtual plug and socket depth images, (2) a wrist-mounted depth observation, and (3) a 3D end-effector force measurement. The AGOS depth observation and wrist depth observation are encoded using separate ResNet18 encoders, producing 512-dimensional feature vectors, while force measurements are encoded using an MLP into a 128-dimensional feature vector. The resulting features are concatenated into a 1152-dimensional representation and projected to a 512-dimensional conditioning vector through a linear layer. Conditioned on this fused representation, a diffusion policy predicts an action chunk, where each action consists of a translational displacement ($\Delta x, \Delta y, \Delta z$) and a 6D rotation representation ΔR_{6D} .

We build our policy on top of Diffusion Policy and predict action chunks consisting of end-effector translational and rotational actions. The policy takes three observation modalities as input: (1) wrist depth observations, (2) aligned geometric observations (AGOS), and (3) end-effector force observations.

The wrist depth observation is encoded using a ResNet18 encoder adapted for single-channel depth input. The aligned geometric observation space $\mathcal{G}_t = \tilde{I}_p, \hat{I}_s^t$ is represented as a two-channel depth image obtained by stacking the aligned plug and socket depth observations, and is encoded using a separate ResNet18 encoder. Both encoders produce 512-dimensional feature embeddings. The force observation is processed using a two-layer multilayer perceptron that outputs a 128-dimensional feature vector. The architecture uses separate encoders for wrist observations and AGOS observations to allow the network to learn complementary representations for local insertion state estimation and plug-socket geometric alignment. The implementation follows the architecture shown in our code-base.

The resulting feature embeddings are concatenated and projected into a shared latent representation using a fully connected fusion layer. This latent representation conditions a diffusion-policy action decoder implemented using a Conditional U-Net backbone. During training, the diffusion model is optimized to predict expert action sequences from demonstration data. During inference, the policy

494 generates action chunks using DDIM [43] sampling and executes the first portion of the predicted
495 action sequence before replanning.

496 B.2 Observation Modalities

497 The wrist depth observation provides local visual information about the insertion state and contact
498 region. AGOS provides an occlusion-aware representation of plug-socket geometry and rotational
499 alignment. The force modality provides contact information that is particularly useful during in-
500 sertion failures and recovery behaviors. The combination of these modalities enables the policy to
501 reason jointly about geometry, insertion progress, and contact interactions.

502 For all experiments, AGOS observations are represented as aligned depth images, force observations
503 are represented as three-dimensional force vectors, and action sequences consist of translational and
504 rotational end-effector actions represented using a 3D translation and 6D rotation parameterization.

505 C Procedural Generation Details

506 This appendix provides implementation details of the procedural shape generation pipeline used to
507 construct the pre-training dataset. Figure 3 show insertion interface shapes that can be generated
508 through our 2D Insertion Interface generator.

509 C.1 2D Insertion Interface Generation

510 We first generate a 2D insertion interface profile that defines the cross-sectional geometry shared
511 by both the plug and socket. The profile is constructed from geometric primitives including circles,
512 ellipses, rectangles, rounded rectangles, squares, and rounded squares.

513 To increase geometric diversity, primitives are composed into several layout categories:

- 514 • **Single-part layouts:** a single primitive defines the insertion interface.
- 515 • **Two-part asymmetric layouts:** two primitives with independently sampled sizes and off-
516 sets are merged to form a non-symmetric shape.
- 517 • **Three-part asymmetric layouts:** a central bridge component is combined with indepen-
518 dently sampled left and right components.
- 519 • **Clover layouts:** four overlapping primitives are arranged around a common center to gen-
520 erate shapes with higher-order rotational symmetry.

521 We randomize the layout type, primitive type, primitive dimensions, geometric transformations
522 (scaling and rotation), and part offsets within predefined ranges. These variations produce inser-
523 tion interfaces with diverse shapes and rotational alignment characteristics. Primitive generation
524 parameters are summarized in Table 3.

525 Figure 4 provides an example of the procedural generation process, including layout selection, primi-
526 tive sampling, random geometric transformations, and boolean union for constructing the final in-
527 sertion interface geometry.

528 C.2 Shape Filtering and Normalization

529 After shape composition, the resulting polygon is randomly rotated and optionally smoothed using
530 morphological operations. Shapes are discarded if they violate geometric quality constraints.

531 Specifically, we reject shapes with:

- 532 • Aspect ratio greater than 3.6,
- 533 • Area below a minimum threshold.

534 Those shapes are rejected because they are not suitable for insertion tasks.

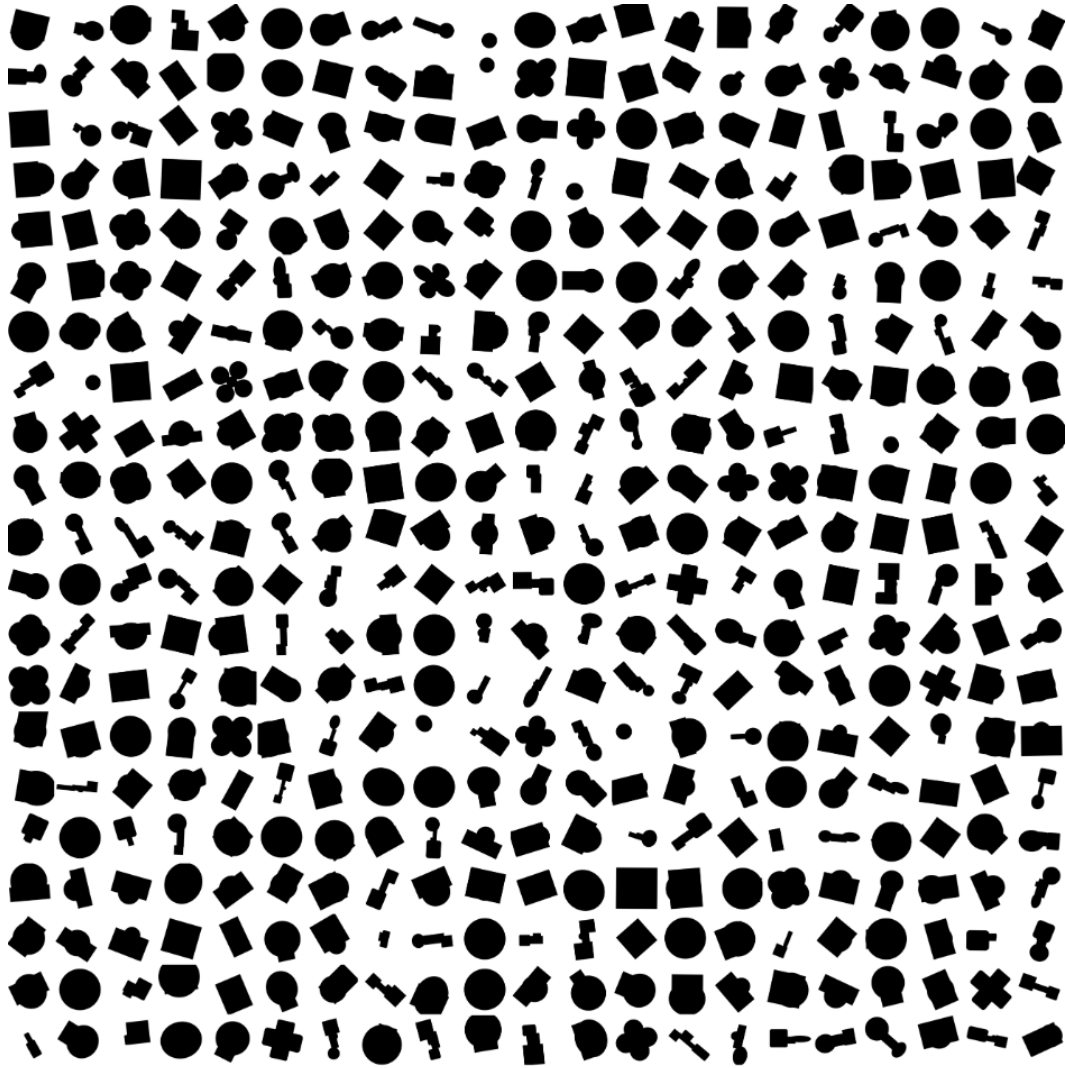


Figure 3: Examples of procedurally generated insertion interface geometries. The generator composes insertion interfaces from primitive geometric components with randomized scale, position, and orientation, producing diverse rotational alignment constraints for training and evaluation.

C.3 Plug Generation

The generated 2D profile is extruded along the insertion axis to form the plug geometry.

The insertion portion of the plug has a height of 30mm. A cylindrical grasping post is attached above the insertion geometry to facilitate robot grasping. The grasp post has:

- Height: 35mm,
- Radius: 4mm.

A short blending section is inserted between the insertion geometry and grasp post to avoid sharp geometric transitions.

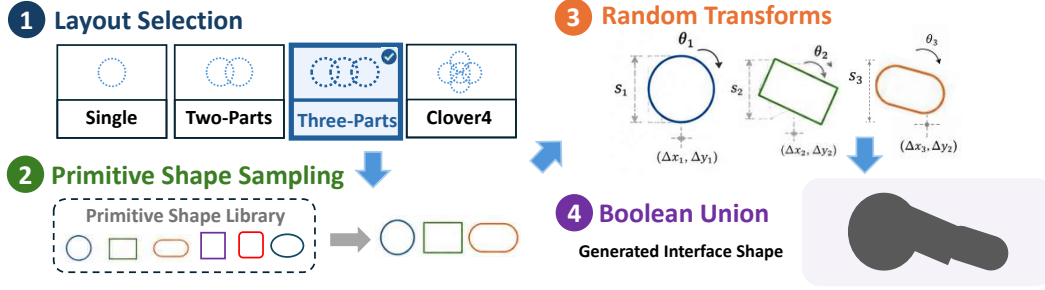


Figure 4: Procedural generation pipeline used to generate diverse insertion interface geometries. A layout template is first selected to determine the arrangement of primitive slots. Primitive shapes are then independently sampled from a predefined library and randomly transformed through scaling, rotation, and translation. The transformed primitives are subsequently merged through boolean union to form a single insertion interface profile. This process enables systematic generation of geometries with diverse shapes, symmetries, and rotational alignment requirements. In the illustrated example, a three-part layout is selected, followed by sampling a circle, rectangle, and ellipse primitive. After applying random geometric transformations, the primitives are merged to produce the final insertion interface geometry.

Table 3: Sampling parameters used for procedural generation of insertion interface geometries. All geometric parameters are defined in normalized shape coordinates and are subsequently scaled when generating the final mesh.

Category	Parameter	Range
Layout	Layout type	Single, Two-Part, Three-Part, Clover4
	Layout probability	{0.10, 0.50, 0.30, 0.10}
	Part vertical offset	$[-0.38, 0.38]$
Primitive Shape	Primitive type	Circle, Ellipse, Square, Rectangle, Rounded Square, Rounded Rectangle
	Circle radius r	$[0.45, 0.95]$
	Ellipse radii (r_x, r_y)	$r_x \in [0.50, 1.00], r_y \in [0.40, 0.90]$
	Rectangle half-dimensions (h_x, h_y)	$h_x \in [0.55, 1.15], h_y \in [0.35, 0.90]$
	Rounded-corner radius	$[0.18, 0.40] \cdot \min(h_x, h_y)$
Transformation	Rotation angle θ	$[0^\circ, 180^\circ]$
	Scale factor	$[0.90, 1.15]$

543 C.4 Socket Generation

544 Figure 5 illustrates the geometric parameters used for socket construction, including the clearance
545 between the plug and socket mating surfaces and the outer margin that defines the thickness of the
546 socket body.

547 Socket geometries are generated from the same insertion profile.

548 To ensure physically feasible insertion, the profile is first expanded using a clearance offset. The
549 resulting cavity profile is then surrounded by an outer wall profile and triangulated using constrained
550 polygon triangulation.

551 The socket parameters are:

- 552 • Socket height: 35mm,
- 553 • Clearance: 0.8mm,
- 554 • Outer margin: 5.5mm.

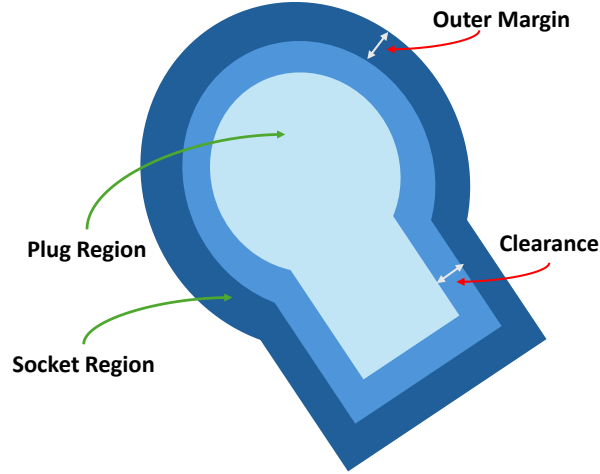


Figure 5: Definition of the clearance and outer margin used in plug-socket geometry construction. Clearance specifies the gap between the plug and socket mating surfaces and determines the insertion tolerance, while outer margin defines the thickness of the socket body surrounding the insertion interface.

The cavity extends approximately 30mm along the insertion direction, matching the insertion depth of the plug.

D Implementation Details for Constructing AGOS

This section provides additional implementation details for constructing the Aligned Geometric Observation Space (AGOS) in both simulation and real-world settings. Although the acquisition process differs between simulation and the real world, both pipelines ultimately produce the same aligned virtual representation used by GeoAdapt.

D.1 Capturing Plug Geometry

AGOS requires geometric observations of both the plug and socket before insertion begins. The socket geometry is obtained from a depth observation captured before insertion and subsequently projected into a virtual image plane. The plug geometry is acquired once before insertion and remains fixed throughout the insertion process.

Simulation. In simulation, the plug geometry is obtained directly from an upward-facing auxiliary depth camera placed beneath the workspace. The plug is positioned above the auxiliary camera while maintaining its insertion orientation. The captured depth image is converted into a point cloud and used to construct the plug observation in AGOS.

Real World. In the real world, we found that placing the plug directly above the depth camera results in unreliable depth measurements due to the minimum sensing range of the camera. To address this issue, we move the plug to an offset viewpoint where the geometry can be accurately reconstructed. The auxiliary camera captures a point cloud of the plug geometry, which is then transformed into the coordinate frame used for AGOS construction.

D.2 Geometric Scale Calibration

To ensure that plug and socket geometries are represented at consistent scales, we perform geometric scale calibration before projecting the observations into the aligned virtual image plane.

579 **Simulation.** In simulation, the plug observation is captured from a physical auxiliary camera,
 580 while the socket observation is rendered from a virtual camera placed above the socket geometry. Let
 581 d_p denote the distance between the auxiliary camera and the plug tip, estimated using the minimum
 582 valid depth value in the plug observation. Since the plug is positioned upright above the auxiliary
 583 camera, the closest visible point corresponds to the bottom tip of the plug. Let d_s denote the height
 584 of the virtual camera above the socket geometry. The plug observation is scaled according to the
 585 ratio d_p/d_s before projection into AGOS.

586 **Real World.** In the real world, we reconstruct the plug geometry from the captured point cloud and
 587 estimate the plug pose using the known end-effector pose. A virtual camera is then placed below the
 588 reconstructed plug geometry using the same camera intrinsics and projection configuration used for
 589 the socket observation. Both the plug and socket point clouds are subsequently projected into aligned
 590 virtual image planes, producing geometrically consistent observations for AGOS construction.

591 **D.3 Aligned Virtual Projection**

592 After geometric acquisition and scale calibration, both plug and socket geometries are projected
 593 into a shared virtual image plane. The virtual camera used for the plug observation shares the same
 594 intrinsics as the virtual camera used for the socket observation. As a result, the projected plug and
 595 socket observations are geometrically aligned and represented at consistent scales.

596 During insertion, the plug observation remains fixed while the socket observation is updated accord-
 597 ing to the current gripper yaw rotation. This produces the aligned geometric observation space which
 598 preserves rotational alignment information throughout insertion despite severe visual occlusion.

599 **E Recovery-oriented Demonstration Collection**

600 We collect demonstrations using a scripted insertion controller in Isaac Gym. Unlike assembly-by-
 601 disassembly approaches, we directly execute forward insertion trajectories, allowing demonstrations
 602 to capture realistic insertion contacts and force responses.

603 **E.1 Nominal Insertion Policy**

604 At each control step, the controller computes the translational error between the current plug position
 605 and the target insertion position, using the privileged knowledge in simulation. The insertion action
 606 is determined according to the lateral alignment error. When the lateral error is large, the controller
 607 prioritizes corrective motion in the horizontal plane. As the lateral error decreases, the controller
 608 gradually increases motion along the insertion axis and eventually performs downward insertion
 609 once the plug is sufficiently aligned with the socket.

610 To generate rotational alignment behavior, we additionally compute the relative yaw error between
 611 the current plug orientation and the target insertion orientation. The controller incrementally rotates
 612 the plug toward the target orientation using small yaw correction steps during insertion.

613 **E.2 Recovery-oriented Demonstrations**

614 To expose the policy to contact-rich failure states, we perturb the target insertion pose before inser-
 615 tion begins. For each trajectory, we sample translational offsets in the horizontal plane and rotational
 616 offsets around the insertion axis. The perturbed pose is maintained during the initial stage of inser-
 617 tion, causing the plug to approach the socket under controlled misalignment.

618 After a fixed number of control steps, the controller switches back to the nominal target pose and
 619 executes corrective recovery behavior. This produces demonstrations containing both failure states
 620 and successful recovery motions.

Algorithm 1 Recovery-oriented Demonstration Collection

```
1: for each insertion episode do
2:   Construct aligned geometric observation space  $\mathcal{G}_0$ 
3:   Sample perturbations  $\delta p \sim \mathcal{P}_{\text{trans}}, \delta \psi \sim \mathcal{P}_{\text{yaw}}$ 
4:   Initialize insertion pose  $x_0 = x_g \oplus (\delta p, \delta \psi)$ 
5:   while insertion not complete do
6:     Update aligned observation space  $\mathcal{G}_t$ 
7:     Execute scripted insertion action  $a_t$ 
8:     if  $|\hat{f}_t^z| > \tau_f$  then
9:       Suppress actions along the insertion axis
10:    Allow lateral corrective motions to dominate the action update
11:    end if
12:    Store  $(I_t^w, \mathcal{G}_t, \hat{f}_t, a_t)$  into  $\mathcal{D}_{\text{pre}}$ 
13:  end while
14: end for
```

621 We generate two categories of recovery trajectories. Easy recovery trajectories use rotational per-
622 turbations between 5° and 90° , while hard recovery trajectories use perturbations between 91° and
623 179° .

624 E.3 Force-triggered Recovery Behavior

625 To increase the frequency of recovery behaviors, we monitor force measurements from the fingertip
626 force sensors during insertion. When a rapid increase in insertion force is detected, indicating a
627 likely collision or jamming event, the controller temporarily reduces downward insertion motion
628 and prioritizes lateral corrective actions.

629 Specifically, the insertion step size along the insertion axis is reduced while horizontal corrective
630 motions continue. Once the contact force decreases, the controller resumes normal insertion be-
631 havior. This mechanism generates demonstrations containing realistic recovery behaviors under
632 contact-rich failures and improves the diversity of recovery motions observed during training.

633 E.4 Data Collection Algorithm

634 Algorithm 1 shows how we collect collected pretraining datas trajectories.

635 F Zero Shot Performance for Simulation Experiments

636 Table 4 reports the zero-shot success rates on unseen insertion geometries. GeoAdapt achieves the
637 highest average success rate of 48.2%.

Table 4: Zero-shot Success Rates on Unseen Geometries. Tasks are ordered by increasing rotational alignment difficulty from left to right. We report success rates (%) on eight unseen socket and plug geometries in simulation, evaluated over 100 rollouts per geometry. GeoAdapt achieves the highest average success rate compared to its ablated variants and the GeoAdapt-Automate baseline.

Methods									Average
GeoAdapt	63	70	58	62	37	37	24	35	48.2
w/o Force	31	77	54	46	31	35	17	26	39.6
w/o AGOS	51	55	37	53	25	32	8	28	36.1
w/o Scale Calibration	60	60	55	56	33	20	11	32	40.9
w/o Rot. Augmentation	66	68	61	49	25	24	21	25	42.4
GeoAdapt-Automate	13	15	10	20	3	5	0	8	9.3

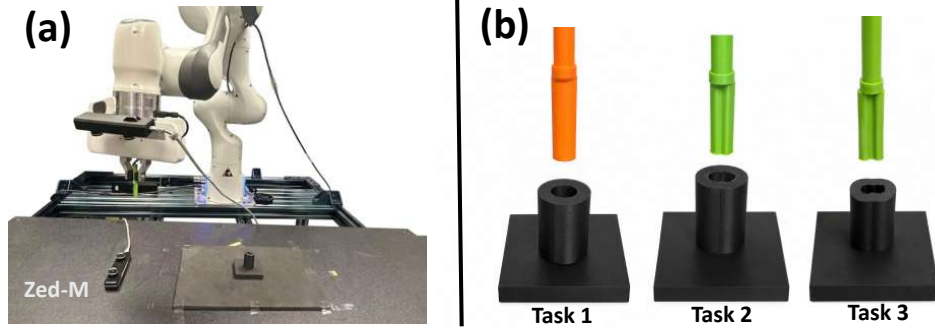


Figure 6: Real-world experimental setup and evaluation tasks. (a) A Franka Panda robot equipped with a wrist-mounted ZED Mini depth camera and an auxiliary ZED Mini depth camera positioned beneath the workspace. The auxiliary camera captures plug geometry observations for constructing the aligned geometric observation space (AGOS). (b) Three 3D-printed plug-socket geometries used for real-world evaluation. These geometries are excluded from the training dataset and are ordered from left to right by increasing rotational alignment difficulty.

G Real World Experiments Setup

See Figure 6 for the real-world experimental setup. We use a Franka Panda robot equipped with a wrist-mounted ZED Mini depth camera and an auxiliary ZED Mini depth camera positioned beneath the workspace. The auxiliary camera captures plug geometry observations for constructing the aligned geometric observation space (AGOS). Before each insertion trial, the robot first translates the gripper from its initial pose to a position above the auxiliary camera to capture the initial plug virtual observation. The robot then returns to the original pose and executes the insertion policy. We evaluate GeoAdapt on three unseen 3D-printed plug-socket geometries that are excluded from the training dataset and ordered by increasing rotational alignment difficulty.

H GeoAdapt can Recover from Failure in Real World Experiments

GeoAdapt can actively recover from failure states caused by rotational misalignment during real-world execution, illustrated in Figure 7.

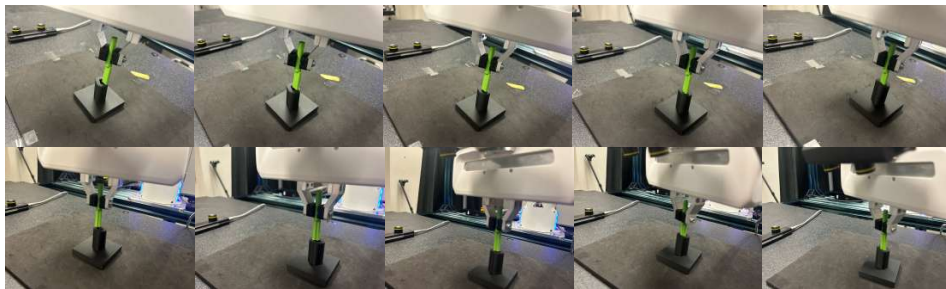


Figure 7: Real-world insertion sequence on an unseen insertion geometry. Starting from a rotationally misaligned configuration, the plug becomes stuck at the socket opening due to geometric mismatch. GeoAdapt actively executes corrective rotational motions to recover from the failure state, progressively aligning the plug with the socket geometry before successfully completing the insertion